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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	牛马来西亚
Selected Domain	Domain2
Selected SDG Track (1–17)	SDG12
Project / Solution Name	PrepWise（Demand Forecasting and Meal Preparation Planning for Small Restaurants）
Name (Leader)	LAN DONGAO
Email & Phone (Leader)	+60 1123318006
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	WEI SHANGAO， SHI JIAHUAN ,CHEN YAN,CUI LIUYANG

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

1.Small and medium-sized restaurants face the daily challenge of preparing the right amount of food efficiently, accurately, and profitably. Restaurant owners and employees must determine how much food to prepare each day to meet customer demand while controlling costs.

2.Manual food preparation planning is time-consuming, inefficient, and prone to human error. Preparing too much food leads to unnecessary ingredient costs and food waste, while preparing too little can result in stock shortages, lost sales, and customer dissatisfaction. These problems reduce profit margins and place additional financial pressure on restaurants, potentially disrupting their normal operations.

3.If these challenges are not addressed, restaurants may continue to experience avoidable financial losses and operational inefficiencies. Therefore, an effective solution is needed to improve daily food preparation decisions. By addressing this problem, restaurants can reduce costs, minimize food waste, improve operational efficiency, and maximize profitability while meeting customer demand.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Our proposed application is an AI-powered smart meal preparation planning system designed to help small and medium-sized restaurants transition from experience-based daily meal preparation decisions to data-driven planning.

The system uses an XGBoost machine learning model to forecast next-day sales for individual menu items based on historical sales data, weather conditions, calendar information, and other relevant factors. It then

converts predicted demand into ingredient requirements and analyzes them against recipe data and existing inventory to generate data-driven meal preparation and purchasing recommendations.

The application provides restaurant owners and staff with a simple, user-friendly interface to view meal preparation recommendations and relevant operational information. It also records actual sales data and evaluates the prototype using measurable indicators, including food waste reduction, meal preparation accuracy, and time saved in daily planning. By comparing actual sales with predicted demand, the system can assess forecasting performance and support continuous improvement through regular model retraining.

However, the final decisions regarding meal preparation and daily operations remain under the control of the restaurant manager or designated staff, ensuring that human judgment remains an essential part of the decision-making process.

We plan to apply technical skills gained from the SDC Phase 1 workshops, particularly database architecture for storing and managing sales, recipe, and inventory records.

By combining machine learning with structured data management, our solution aims to improve meal preparation planning and purchasing decisions, reduce avoidable food waste, save staff time, and promote more sustainable restaurant operations.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

| Component | Technology and purpose |

|---|---|

| Frontend | HTML, CSS and JavaScript for data upload, forecast settings, preparation plans and charts |

| Backend | Python 3.12, FastAPI and Uvicorn for application services and REST APIs |

| Data processing | Python CSV utilities and openpyxl for validating CSV/XLSX sales records |

| Forecasting | NumPy-based Ridge regression, XGBoost and a historical baseline, compared through time-based validation |

| Database | SQLite for dish information, synthetic sales records and saved forecast snapshots |

| Model storage | JSON artifacts containing trained models, validation results and data fingerprints |

| Hosting | Replit for the deployed frontend and backend |

| Version control | GitHub for source code, documentation and collaboration |

The system connects API design, database management and cloud deployment into one workflow. Users upload team-generated synthetic sales data through the interface. FastAPI validates and stores the records, invokes the forecasting module, and returns suggested preparation quantities. Users can review results, export plans and record synthetic actual sales for comparison.

```mermaid
flowchart LR
    U[Restaurant operator] --> UI[Web interface]
    subgraph Replit cloud
        UI --> API[FastAPI REST APIs]
        API <--> DB[(SQLite database)]
        API <--> ML[Baseline / Ridge / XGBoost]
        ML <--> JSON[JSON model artifacts]
```

```
    API --> UI
  end
  UI --> R[Review plan, export CSV and view feedback]
```

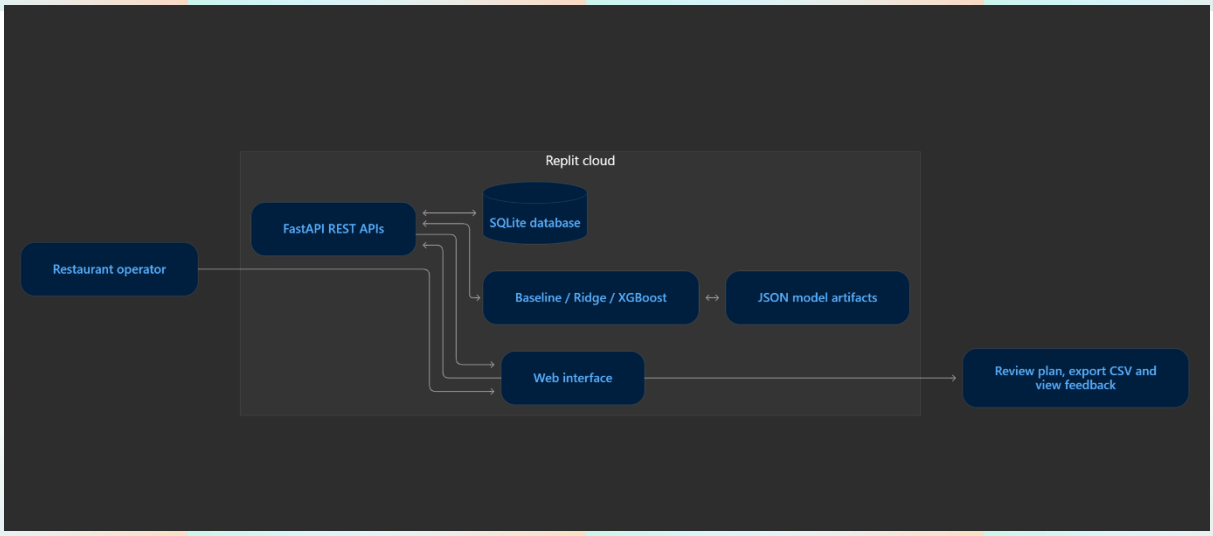
Use of official sponsor APIs or cloud platforms

Yes. We use Replit, identified in the organizer-linked submission materials, as the cloud platform for hosting the functional prototype:

https://sdc-hackathon-2026--lansoar000.replit.app/

The current application does not call a sponsor-provided generative-AI API. Its forecasting runs on CPU using the methods listed above. ChatGPT/Codex and Replit Agent assisted development and documentation; they are distinct from APIs embedded in the application.

RunPod integration has not yet been implemented or verified, so it is not claimed as a completed component.



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied the design-thinking methodology by focusing on the practical challenges faced by small and medium-sized restaurants, particularly the reliance on experience-based meal preparation, over-purchasing, and avoidable food waste.

Empathize and Define: We identified the need for a simple and practical tool that helps restaurant owners and staff make better daily preparation decisions using sales and inventory data.

Ideate: We proposed an AI-powered solution using XGBoost to forecast next-day demand and convert predictions into meal preparation and ingredient purchasing recommendations.

Prototype: We plan to develop an interactive dashboard that displays sales forecasts, inventory levels, and preparation suggestions. A structured database will manage sales, recipe, and inventory records, while XGBoost will support data processing and prediction.

Test and Improve: We will evaluate the prototype using measurable indicators, including food waste reduction, meal preparation accuracy, forecasting error, and time saved in daily planning. Feedback from potential users will guide improvements to usability and functionality.

By Phase 2, we aim to deliver a functional, deployed prototype with an accessible interface, integrated data management, and a measurable evaluation process. This iterative approach will help ensure that our solution is technically feasible, user-friendly, and suitable for real-world restaurant operations.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target12.3 "By 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses." Our AI-powered restaurant demand forecasting and inventory optimization system supports this target by using XGBoost to predict next-day food demand, calculate ingredient requirements, and generate data-driven purchasing recommendations. By aligning ingredient purchases with predicted demand and existing inventory, the system aims to reduce over-purchasing, prevent avoidable food waste, and improve resource efficiency in small and medium-sized restaurants.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	In our simulated testing environment, the baseline ingredient waste rate for traditional experience-based meal preparation is 25%. Our AI-powered meal preparation planning system aims to reduce this rate to 10% by improving demand forecasting and preparation planning. These figures are based on simulated testing and will require validation through real-world restaurant trials.
Core Measurable KPI *The specific unit you will track*	Kilograms of food waste avoided per week, measured by comparing actual food waste before and after implementing

Metric Category	Your Response
e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.	the AI-powered demand forecasting and inventory optimization system. Percentage reduction in excess ingredient purchases, measured by comparing the quantity of ingredients purchased beyond actual demand before and after implementing the AI-powered demand forecasting and inventory optimization system.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Achieve a 15% reduction in food waste by the final prototype presentation, measured by comparing food waste per week before and after implementing the AI-powered demand forecasting and inventory optimization system under comparable operating conditions.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
FastAPI,	Backend framework,	MIT,
Pydantic,	Data validation library,	MIT,
Uvicorn,	ASGI server,	BSD-3-Clause,
Starlette,	Web framework,	BSD-3-Clause,
openpyx,	Excel processing,	MIT,
NumPy,	Numerical computing library,	BSD-3-Clause; additional
XGBoost,	Machine learning library,	bundled notices apply ,
python-multipart,	File upload parsing library,	Apache-2.0,Apache-2.0,
SQLite	Database engine	Public domain

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

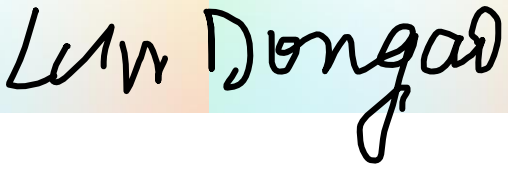
☐ Yes √ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]LAN DONGAO



[Date]24/10/2026